

Conditioning the A/C Technician

A relatively new A/C training system aims to turn A/C technicians into expert diagnosticians.

by Mary James

What do you get when you mix an A/C technician, a little bit of training, and no mentoring? According to John Proctor, principal of Proctor Engineering, you get what you deserve: a plague of poorly operating, energy-wasting A/C systems. That's why he developed CheckMe!, a training and guidance system that combines targeted instruction on A/C essentials, computerized expert systems, and ongoing hand-holding via telephone for field technicians (see "Fast Checks for Refrigerant Charge?," p. 18).

It's a quiet revolution he's fomenting in the A/C world. Since the introduction of CheckMe! in 1998, 20,498 customers have received CheckMe! certificates telling them how their A/C units performed on charge and air flow tests, both before and after the technician worked on the unit.

The certificate is the final feedback loop for technicians enrolled in this system, which places equal emphasis on technical accuracy and human interactions. The CheckMe! system starts with a series of trainings that detail how to diagnose and repair charge, airflow, and duct problems. Proctor designed his training program to address these three problems after 13 years of field research revealed how common they are. Over three-quarters of all A/C units perform inefficiently because of problems with ducts, air flow, or charge.

Finesse in the Field

After a couple of hours of classroom training, field training begins. The instructors work on air conditioners



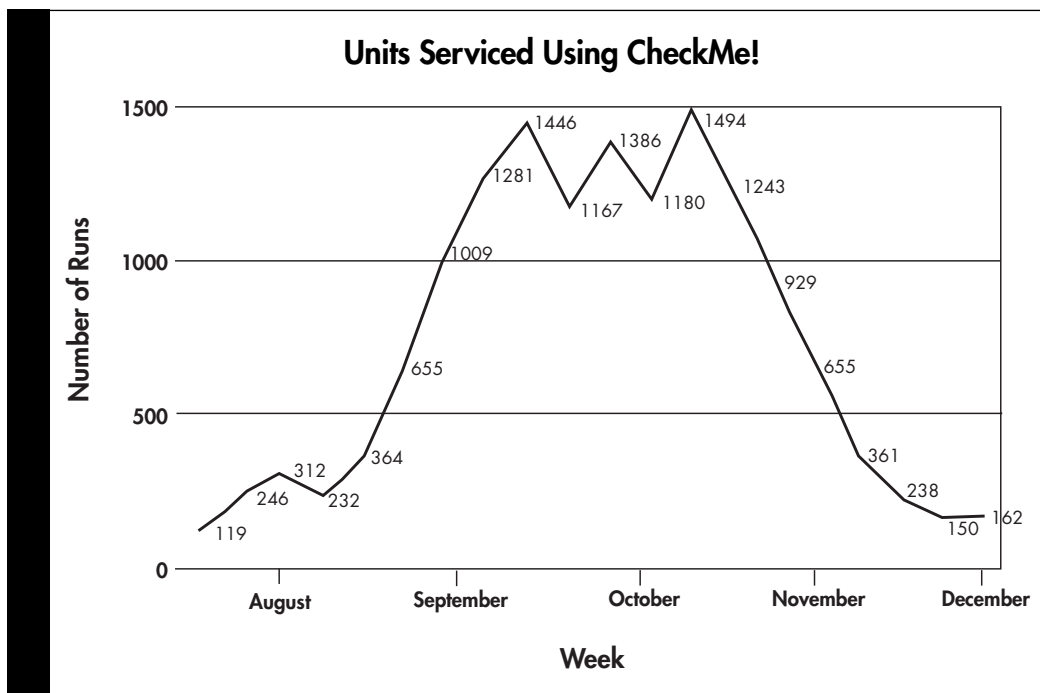
PROCTOR ENGINEERING GROUP

with the technicians until they are reasonably sure that the technicians understand what they need to do and how to do it. "For almost all technicians, checking refrigerant charge by using superheat and really addressing air flow—rather than just checking to see if air is coming out of the register—is completely changing how they usually operate," says Proctor.

The training covers accurate methods for getting needed data, such as the condenser air entering temperature and the return air wet-bulb

temperature, and the shortcuts not to take. An A/C unit has to run for at least 15 minutes before any measurements are taken. "Many technicians think 10 minutes is probably OK," says Proctor, "but then they get into trouble."

With most service calls, technicians might not know or might be able to ignore the fact that they are getting into trouble—but no technician can ignore trouble during a CheckMe! servicing. That's because, after taking the measurements, the technician phones in the data and talks to a live person every time. An



(above) **Figure 1.** In the last half of 2001, 229 technicians used the CheckMe! system to service 6,330 A/C units.
(left) A technician phones in data to a CheckMe! operator.



type of error that is known as conditional occurred?

Most errors fall into the conditional category, and the CheckMe! system will alert a technician to such an error. At that point the operator can refer the technician to a human expert who can walk the technician through a set of possible causes for the error and the needed fixes. "These technicians are never left alone in the field," emphasizes Proctor. "They can always get through to one of us."

The Customer Counts

The third leg of the human interaction tripod that Proctor has set up is the technician's interactions with the customer. The technician knows that every customer will be receiving a CheckMe! certificate after each visit. Knowing this, the technician can't diagnose a refrigerant charge problem, fix that, and then walk away, ignoring any air flow problems. Instead the technician has to explain what is wrong to the customer and bring that customer into the discussion of what steps are needed to

get a fully functioning A/C system.

The customer also receives a postage-paid customer comment card that gets sent to Proctor's office. Seven percent of these cards have been returned, with 96% of these customers rating the service good to excellent. Proctor follows up on every card that delivers a fair or poor rating.

"The CheckMe! system is built around helping technicians do their job right," says Proctor. "We geared the program to provide proactive quality assurance rather than just counting mistakes." Proctor's office tracks each technician's data reporting, and any technicians with high error rates get help. That help may involve ride-

along with a trainer or additional phone consultations.

This intensive training program has won many converts; 229 technicians have signed on in California since last August and have serviced 6,330 A/C units since that time (see Figure 1). Others are calling in their data from Rhode Island, Florida, and several states in the Pacific Northwest. Each one of these calls represents a customer who gets to experience the difference between actually getting an A/C unit fixed and explained to him or her—and just getting the system worked over.

Mary James is the publisher of *Home Energy*.



For more information:

Proctor Engineering Group
818 Fifth Ave., Ste. 208
San Rafael, CA 94901
Tel: (415)455-4700
Web site: www.proctoreng.com